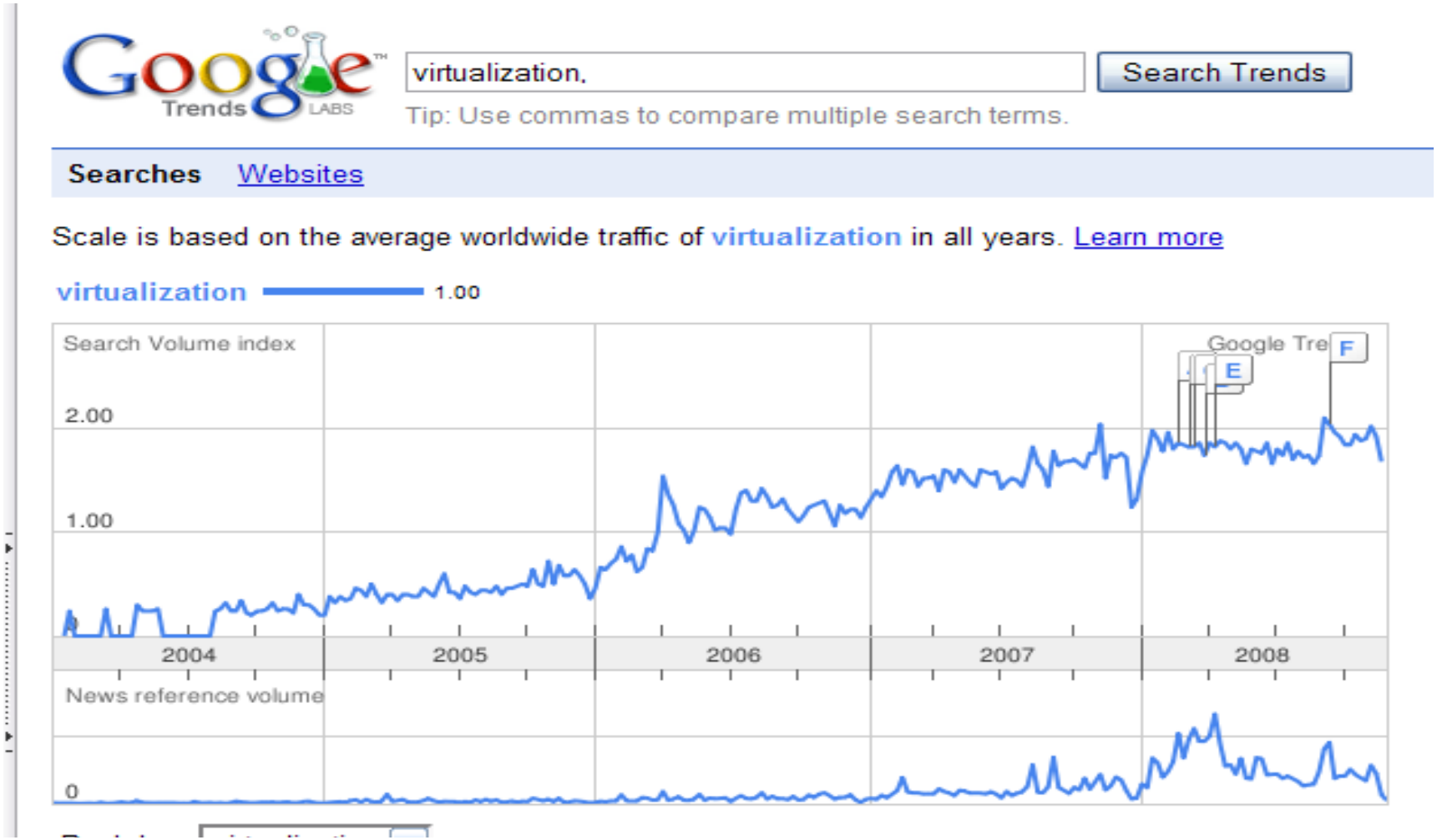
Virtualization: Implications and Aspirations for Middleware

Industry Track Panel

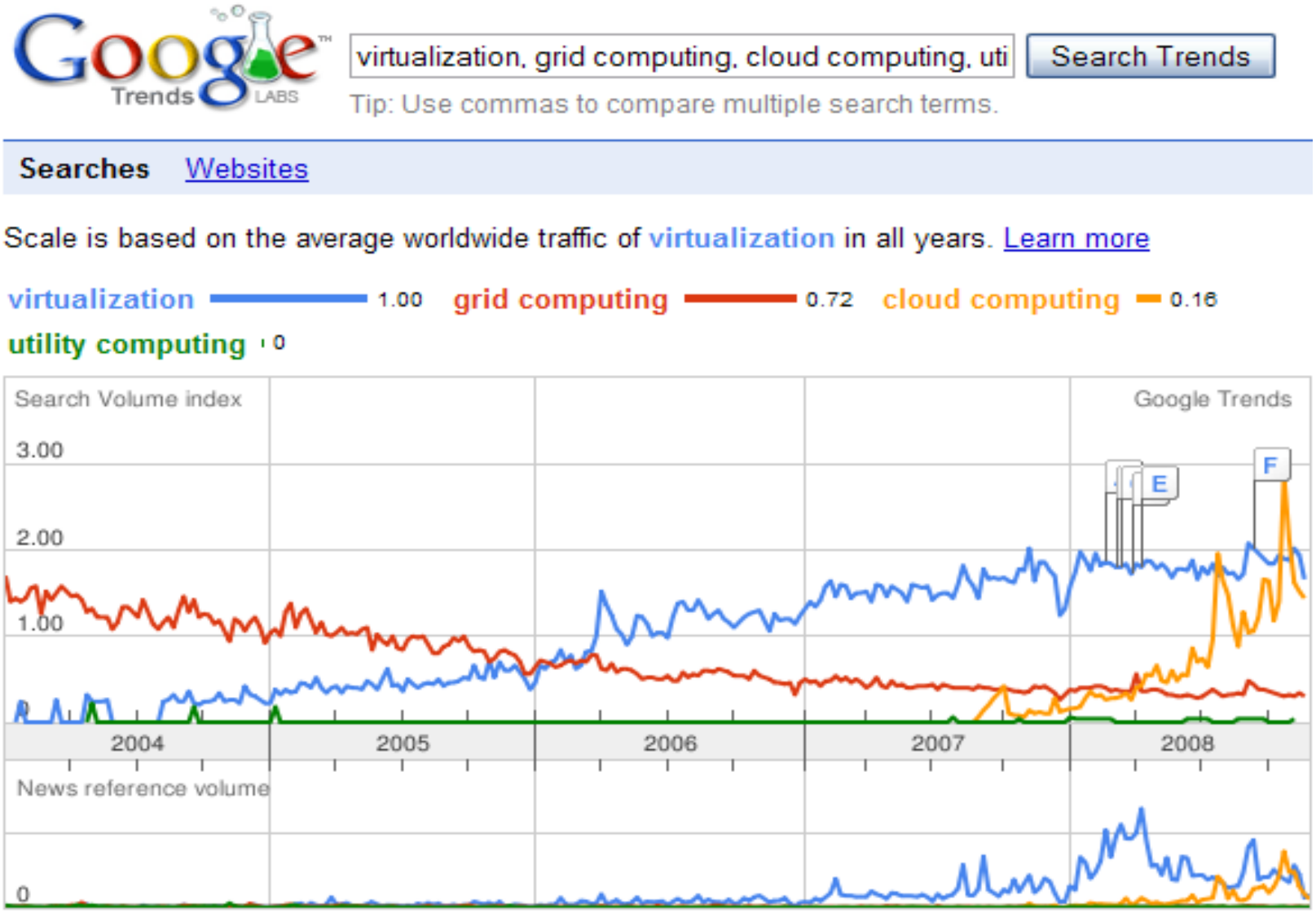
Middleware Conference

4 December 2008

“Obligatory” Trends Slide



Buzzword-competitive



Who We Are



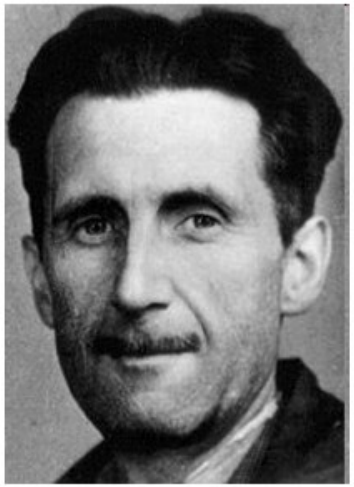
Fred Douglis
IBM T.J. Watson Research Center
Current interests: Stream Computing
Related background: Operating Systems, File systems, mobility

Who We Are



Professor **Roy Campbell**
University of Illinois at Urbana-Champaign
Current interests: Cloud Computing
Related background: Operating Systems, Security, CyberInfrastructure

Who We Are



Ernie Cohen (looks like George Orwell)
Verification architect for the Microsoft Hypervisor (core of Hyper-V)
Current interests: verification of system software
Related background: security, distributed systems, middleware

Who We Are



Prof. **Atul Prakash**
University of Michigan
Current interests: security and privacy, middleware for emerging applications
Related background: cloning file system for virtual servers (Prism),
using virtual machines to detect malware

Who We Are



Prof. **Liuba Shrira**
Professor at Brandeis University, Affiliated Researcher at CSAIL/MIT
Current interests: time travel in storage, security
Related background: storage replication, software upgrades,
spelunking (kind of time travel)

Who We Are





Fred Douglis
IBM T.J. Watson Research Center
Current interests: Stream Computing
Related background: Operating Systems, File systems, mobility



Professor **Roy Campbell**
University of Illinois at Urbana-Champaign
Current interests: Cloud Computing
Related background: Operating Systems, Security, CyberInfrastructure

Ernie Cohen (looks like George Orwell)
Verification architect for the Microsoft Hypervisor (core of Hyper-V)
Current interests: verification of system software
Related background: security, distributed systems, middleware





Prof. **Atul Prakash**
University of Michigan
Current interests: security and privacy, middleware for emerging applications
Related background: cloning file system for virtual servers (Prism),
using virtual machines to detect malware

Prof. **Liuba Shrira**
Professor at Brandeis University, Affiliated Researcher at CSAIL/MIT
Current interests: time travel in storage, security
Related background: storage replication, software upgrades,
spelunking (kind of time travel)

Themes & Questions

- What are we gaining with virtualization? What are we sacrificing?
- What new functionality is enabled?
- Where will virtualization be in 1,5,10 years
- What advice would you give Vmware, Xen, etc? What are we missing?

Panel Format

- Each panelist has about 8 minutes to cover these topics
- This leaves close to ½ hour for questions
  - Perhaps one question after each panelist finishes

Virtualization: Implications and Aspirations for Middleware

Roy Campbell
University of Illinois at Urbana-Champaign

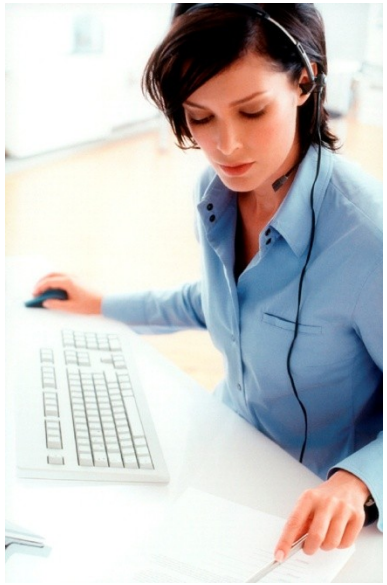


Questions



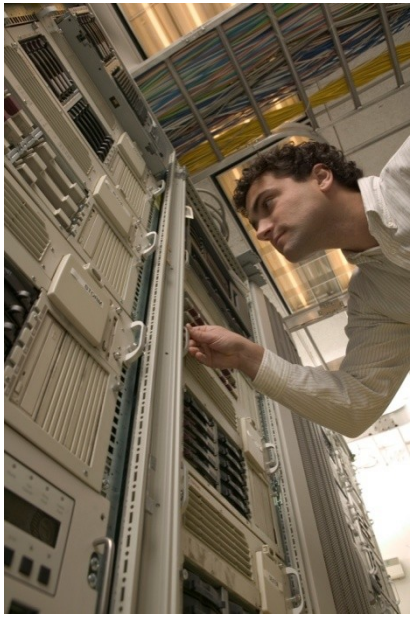
- What do we gain from virtualization?
  - Flexible growth and reuse of systems and configurations. Cost sharing. Reliability and availability. Shared improvements and innovations
- What do we lose?
  - Simplicity, supply side independence, common mode failure
- What key features are missing?
  - Cross cutting performance guarantees
  - Security, privacy

Outsourcing the Academic Infrastructure



- Virtualized resources:
  - Research: shared huge data sets, collections, scalable processing, multiprocessing
  - Education: highly available, communication, instructional resources (email, calendar, courses,...)

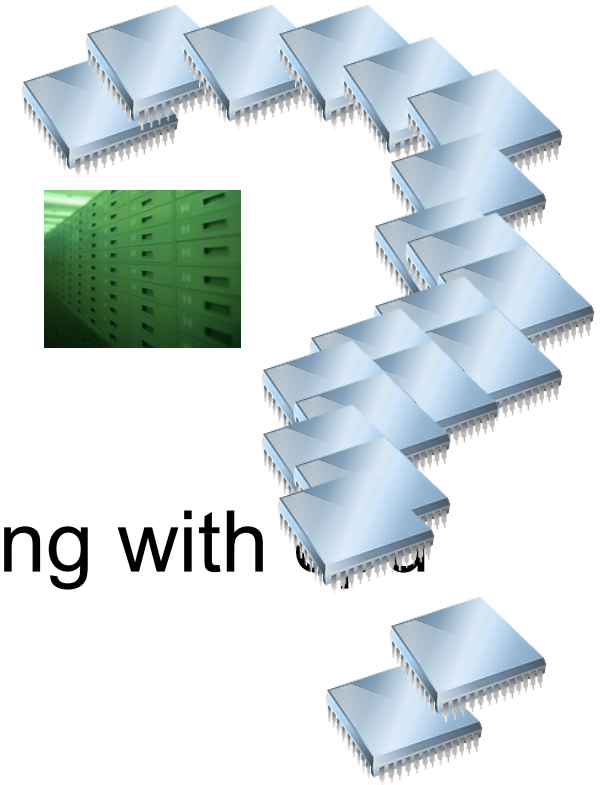
Architectures to support virtual provisioning



- Vlans, vwans, virtual OS, virtual storage
- Location modularities: chip, rack, cluster, room, city, network, provider
- Declarative (graphic) languages for virtual provisioning

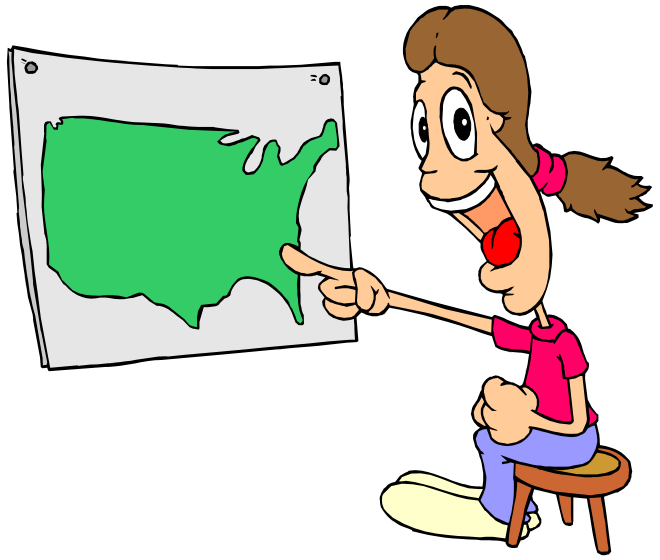
Storage/scheduling virtualized data intensive systems?

- Moving computation or data?
- Network infrastructure
- Disks/storage per node
- CPUs per node
- Can one mix storage scheduling with [illegible] scheduling



Geographic location and virtual provisioning

- how do you take into account geographic location cleanly in virtual provisioning
- archival, fault-tolerance, latency, data gathering, users





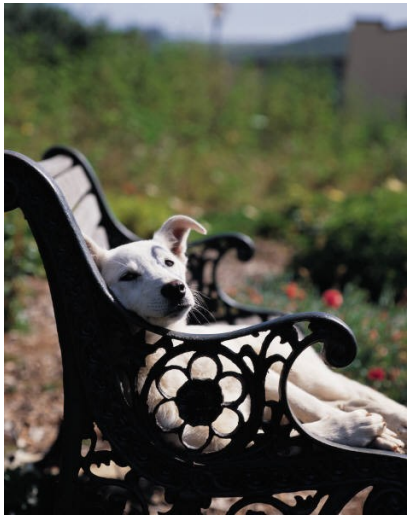
Limits of virtual provisioning

- data intensive computing ≠ supercomputing
- Can virtual provisioning provide both?
- How are cross-level performance attributes offered?



Fair benchmarks for virtualized provisioning/virtual systems

- Availability/performance/reliability....
- Data intensive
- Cloud for Casual Users
- Supercomputing – parallelism?
- Network intensive?



Guarantees versus off peak earnings

- virtual provisioning can guarantee a large corporate its essential on-demand operation peaks
- Sell surplus capacity as cloud provisioning





Trading and selling provisioning?

- Should there be an exchange market for trading and selling virtual provisioning?
- Large finance companies, search companies…
- Peaks and troughs

High end network fabrics with intense computation

- Virtualized interactive 3D (optimistically Holographic) video. High end network fabrics with graphic processing studios between remote sites) and 3D/holographic displays



Where will virtualization be in 1,5,10 years

- Mixing data intensive, supercomputing and on demand clouds
- End to end virtualization
- Multi-core virtualization
- Cellular access
- Instant “on”





ernie cohen

why processor virtualization?

- business driver: flexible provisioning
- middleware driver: OS-neutral illities
- a chance to overcome the OS legacy
  - failure to achieve a process abstraction
  - OSs can't interoperate
  - OSs unworthy of trust
- VMs are the new processes
- hypervisors are the new OS kernels
  - potentially trustworthy (2-100 KLOC)
- middleware is the new modular OS

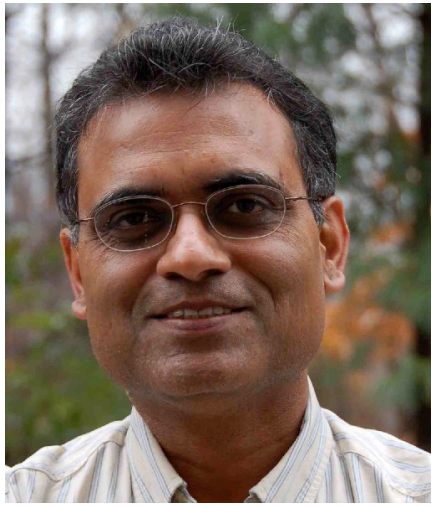
the future

- in 5-10 years, all x64 computing will be hypervised
  - hypervisor-on-metal everywhere
  - microkerneled (no devices in the hypervisor)
  - <5% virtualization overhead
  - (mainly thanks to improved hardware)
- market consolidation
  - a few good, free commodity hypervisors
  - machine migration across vendors
- a rich virtualization ecosystem, with VMs as components
  - small, specialized OSs
  - virtual devices
  - illities
- a path to building trustworthy systems
  - verified hypervisors
  - verification tools for low-level code

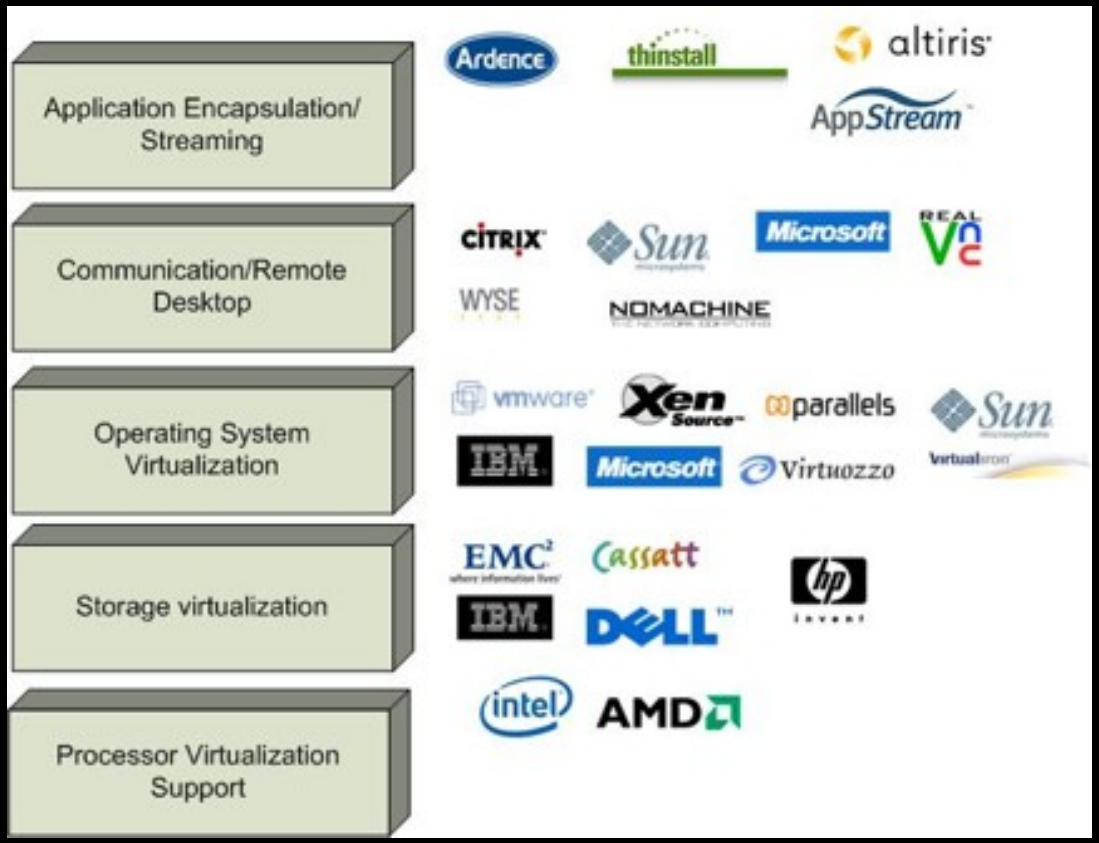
what do we need from the hardware/hypervisor vendors?

- a stable, well-defined platform to build on
  - formal specs
  - strong enough for legacy OSs
  - flexible enough for future hardware
  - buyin from hardware vendors
  - consolidated hypercall API (POSIX for VMs)
- don’t repeat the mistakes of the OS world
  - e.g. don’t try to force the community to sit on top of your favorite new management abstractions
- make hypervisors a free commodity; make your money in the ecosystem

Atul’s Slides



Current Promises

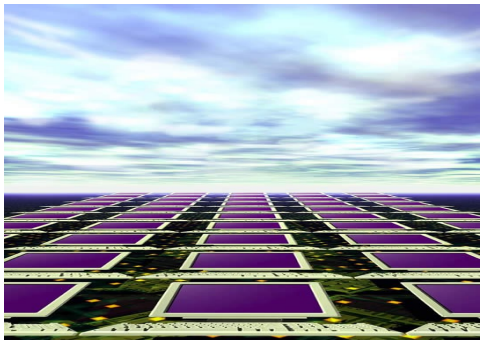


What is missing?

- Maintaining virtual machines: cost of software maintenance cost of virtual PCs not lower: currently, maintained like regular machines
- Dormant virtual machines can pop up and mess up security
- Easier to "steal" the entire virtual PC
- 3D-graphics and I/O passthrough (some hardware support exists from Intel, but software support lagging)

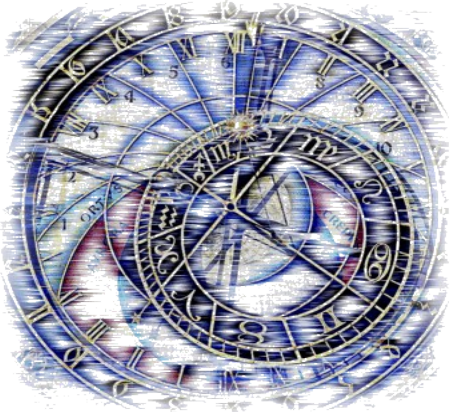
5-10 years from now?

- Rethink storage and file systems to support virtualized environments.
  - Want features of virtualization (snapshots, migration etc.), but security >= standard file systems
- Dormant VM security
  - Patches without requiring online VMs?
  - VM identity management?
- Dynamic graphics and device pass-through



Time travel in the Virtualized Past

Liuba Shrira
Brandeis University



Time travel is interesting!

"The further back you look the further into the future you see" Churchill

Have you ever wondered: have I seen this before, what else happened then, what is different now?

Desktop

Apple's Time machine/Time capsule .. short trips

Storage

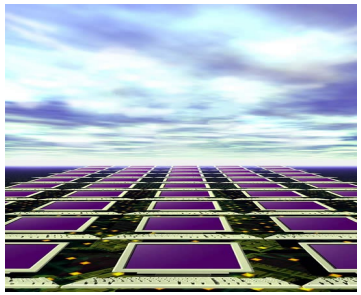
Elephant .. long trips, my area

Literature

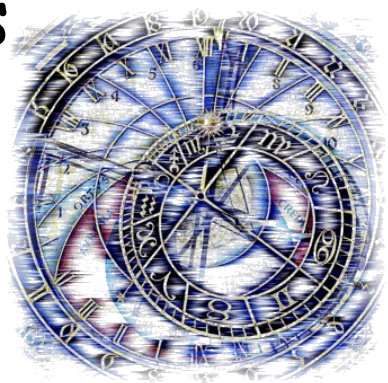
Pensieve from Harry Potter



Revelation:
Time Travel is Virtualization



Much virtualization concerns space - software/hardware layers, platforms, resources, geographical locations



Virtualization in time is

all that and more..

What does it take to TT ?

- Preserve past unobtrusively
  - the stored past can look nothing like the current state
- Reconstitute as needed…
  - so *old* unmodified apps can run on it
  - hence virtualization
  - but also *new* apps
  - not just in pre-planned



"The past is a foreign country, they do things different there.. L.P.Hartley"

- Software evolves
- Hardware gets obsolete
- Privacy norms wane and wax ...

Technical challenges, if you were a graduate student looking for a topic..

- **Virtualization** problems:
  - at what level to capture past states: application, files/DB, storage, VM, or processor - intel 86 - the slowest evolving part!
- **Time travel** problems:
  - how to create past unobtrusively?
  - how far back and how fast you want to travel?
  - on budget? : some states more interesting









Your challenge is a new kind of virtualization problem: how to organize your systems today to enable better time travel tomorrow

And now to Fred's Questions

- What did I gain with virtualization?
  - Can not do much without it..
- What are we sacrificing?
  - More tradeoff than sacrifice:
    - More efficient to integrate past *inside* the virtualization level of present state. *Below* maybe simpler, *above* more complex (see our papers)
- What new functionality is enabled?
  - Can run *retrospective* computations,

  adaptive past for faster and slower travel, GC
- Where will virtualized past be in 1,5,10 years
  - In your brain eventually
- What advice would you give Vmware, Xen, etc?

Open Discussion